

## # Project #2: Wind and Environmental Dynamics

### ## Project Overview

This is the second project in a series of 10 to build a Realistic Video Optimization Agent.

The focus is on detecting and classifying wind effects in generated videos (e.g. hair, curtains, leaves, fabric movement).

### **\*\*Main Learning Goal\*\*:**

Understand Optical Flow computation and basic motion classification using OpenCV.

### ## Final Experiment Results

Video used: Generated hairdryer + hair motion clip

- Total frames analyzed: 966
- jet\_indoor (strong focused wind): 2 frames (0.2%)
- natural\_outdoor (natural scattered wind): 271 frames (28.1%)
- weak\_or\_static (weak or no motion): 693 frames (71.7%)

### **\*\*Observation\*\*:**

The low jet\_indoor percentage indicates that the generated video motion was mostly moderate and scattered rather than strongly concentrated. This is expected in early AI-generated clips.

### ## Core Code Components

1. Video upload and frame extraction
2. Optical Flow calculation using ``cv2.calcOpticalFlowFarneback``
3. Wind type classification based on:
  - Average motion magnitude (``avg_mag``)
  - Direction variance (``ang_var``)

**Current classification logic:**

```
```python
if avg_mag > 4 and ang_var < 2.0:
    wind_type = "jet_indoor" # strong focused wind
elif avg_mag > 2.5 and ang_var > 1.8:
    wind_type = "natural_outdoor" # scattered natural wind
else:
    wind_type = "weak_or_static" # weak or no motion
```